

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

**MECHANICAL
PROPERTIES OF
FLUIDS**

LECTURE-03 PHYSICS

WAJAHID SIR (Gabbar Sir)



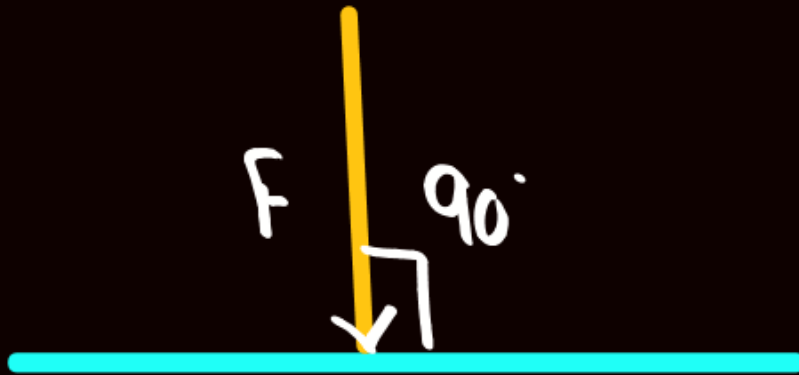
Recap-2

① Application pascal law

- Hydraulic Brake ✓
- Hydraulic lift

② Thrust

- Any perpendicular force is called thrust

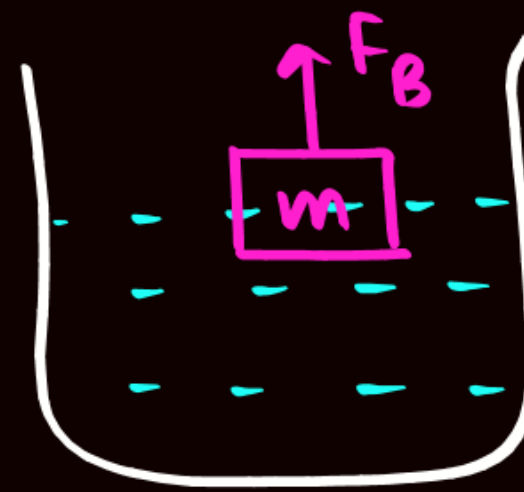


Buoyant force



Archimedes principle

$$F_B = \rho V g$$



Flow

Streamline flow

$$Re < 2000$$

Equation of Continuity

$$a_1 v_1 = a_2 v_2$$

turbulent flow

$$Re \gg \gg 2000$$

Topics To Be Covered

- 1 Energy Possessed by Liquid
- 2 Bernoulli Theorem (2027) / 26 / 25 / 24 / 23 / 22 / 21
- 3 ✓ Application of Bernoulli's Theorem
- 4 ✓ Viscosity
- 5 Stroke Law

RECAP FROM 1ST LECTURE

Fluid

liquid

gas

Density ρ

$$\rho = \frac{m}{V}$$

Pressure

$$P = \frac{F}{A}$$

Atmospheric pressure P_0

Variation of pressure with depth



$$P_B = P_A + \rho gh$$

Pascal Law

Energy Possessed by liquid

(*) All liquids can possess different form of energy depending upon the state of motion

① K.E Kinetic Energy

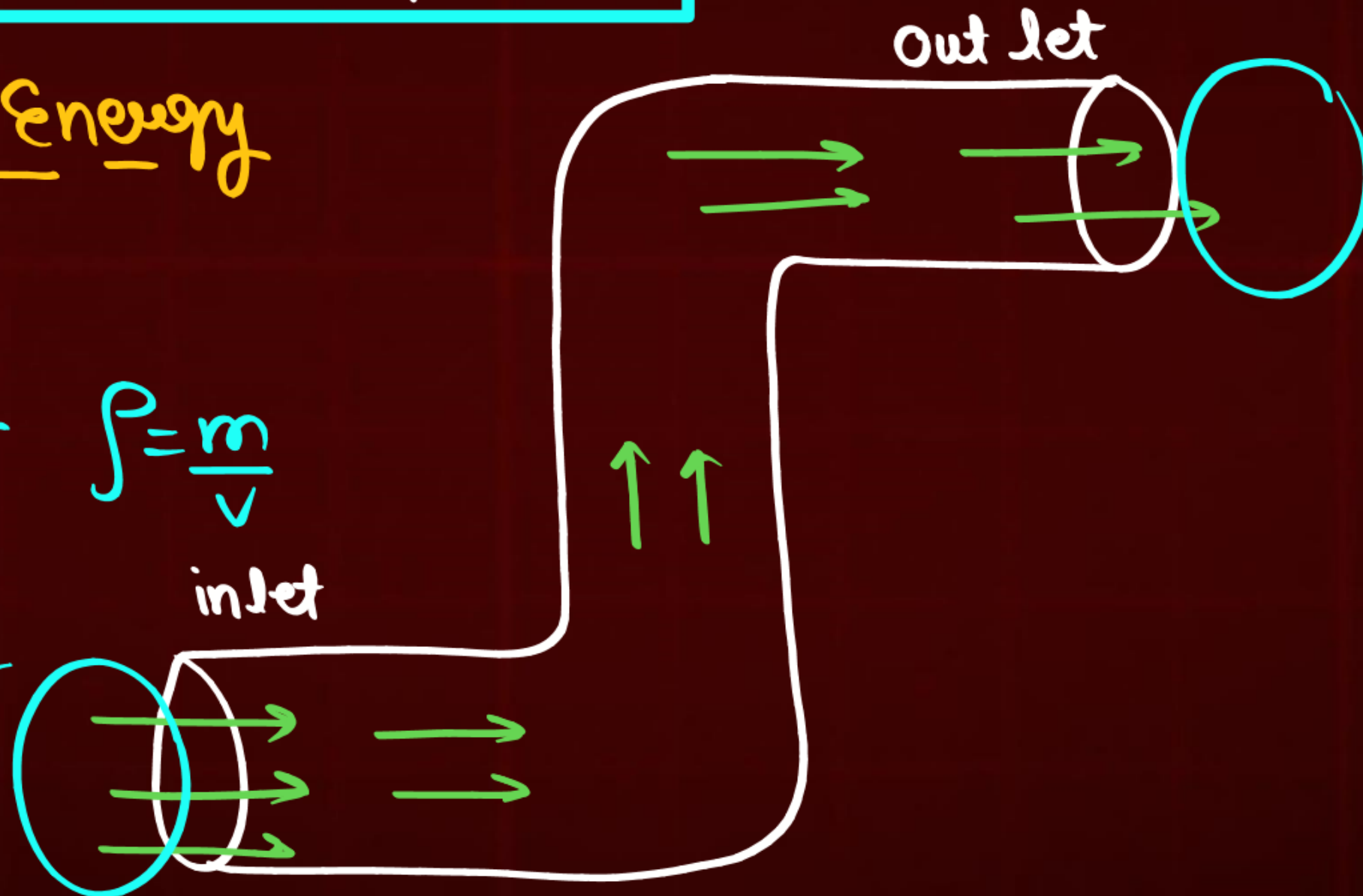
$$K.E = \frac{1}{2}mv^2$$

Kinetic Energy per unit Volume

$$\frac{K.E}{V} = \frac{1}{2} \frac{mv^2}{V}$$

$$\frac{K.E}{V} = \frac{1}{2} \rho v^2$$

$$\rho = \frac{m}{V}$$



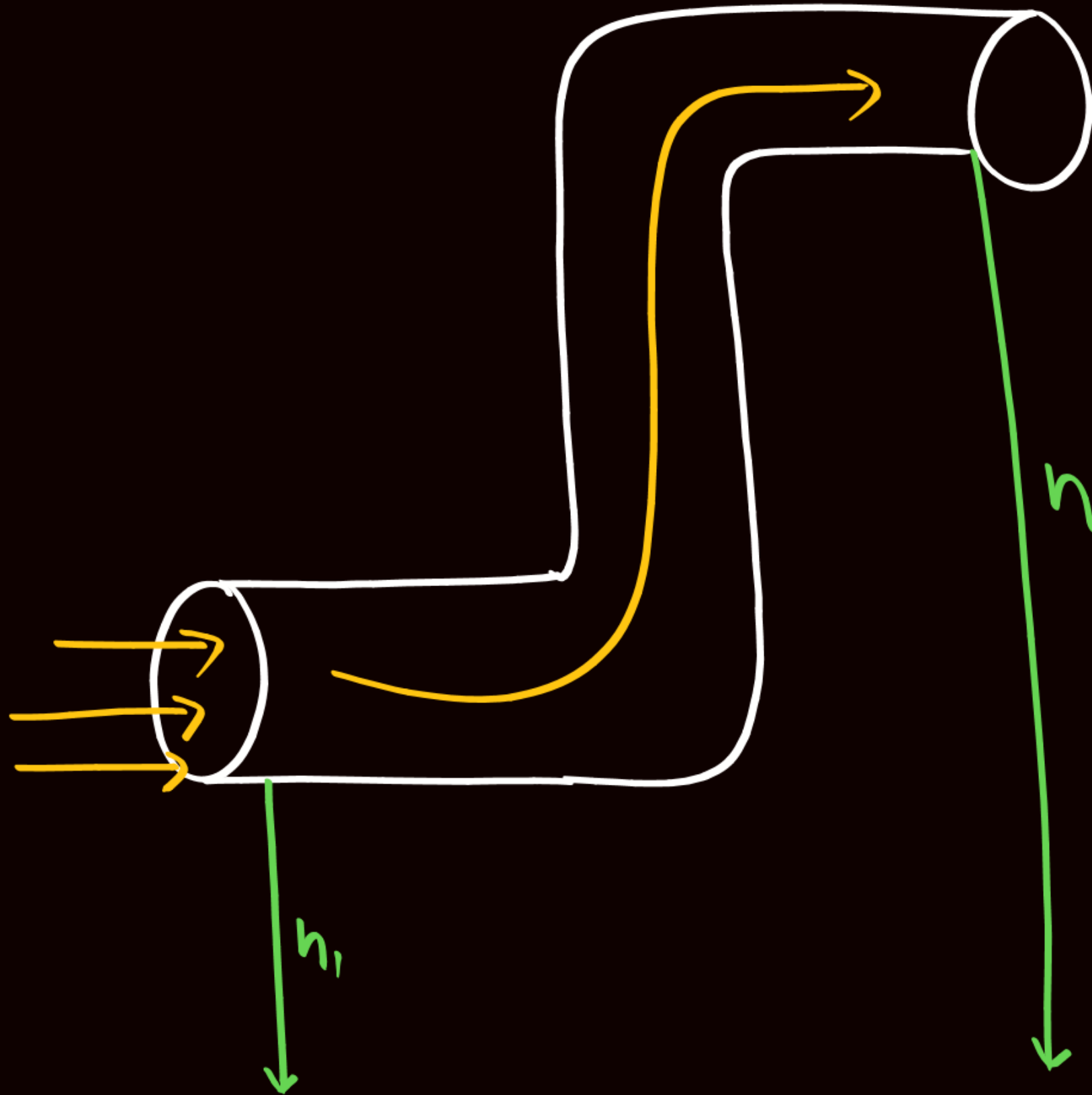
Force
↓
work
↓
Energy

② Potential Energy

$$P.E = mgh$$

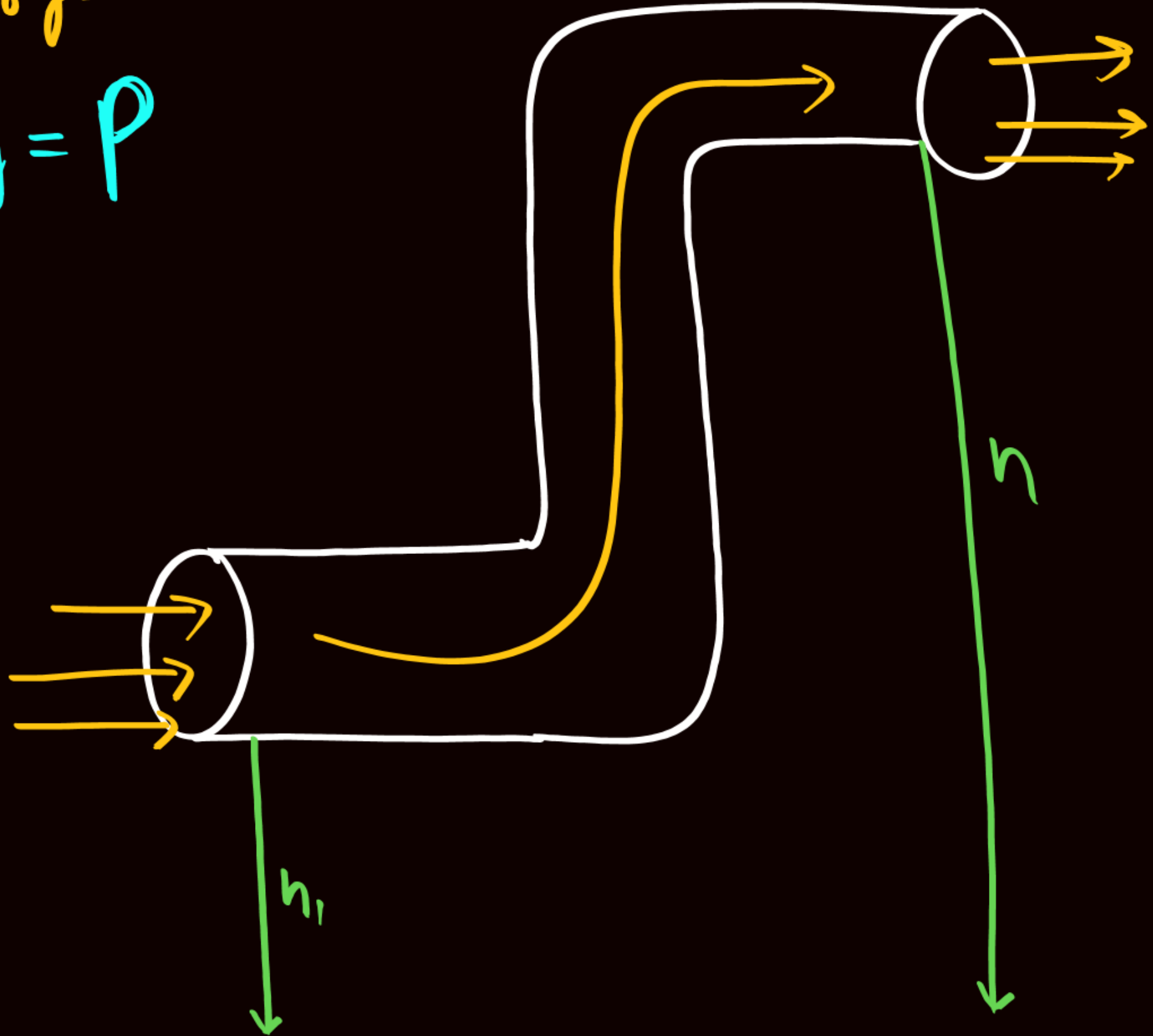
$$\frac{P.E}{V} = \frac{mgh}{V}$$

$$\frac{P.E}{V} = \rho gh$$



③ Pressure Energy

$$\text{Pressure Energy} = P$$



Bernoulli Theorem

V.V.V.V.V.V.V.V. Important. (2027) (05)

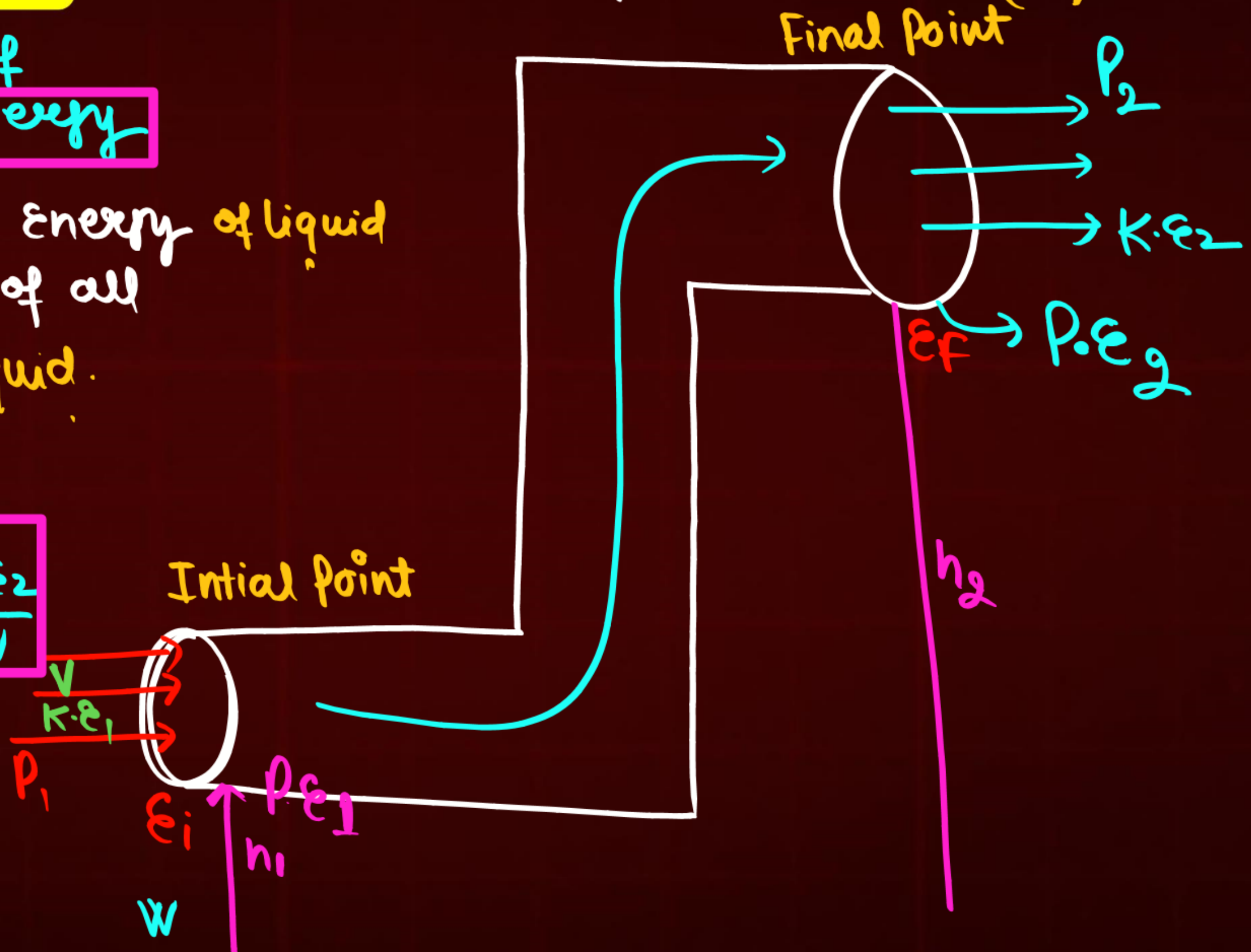


- It is Based on Law of Conservation of Energy

- The Sum of all initial Energy of liquid is equal to the Sum of all final Energy of liquid.

$$\mathcal{E}_i = \mathcal{E}_f$$

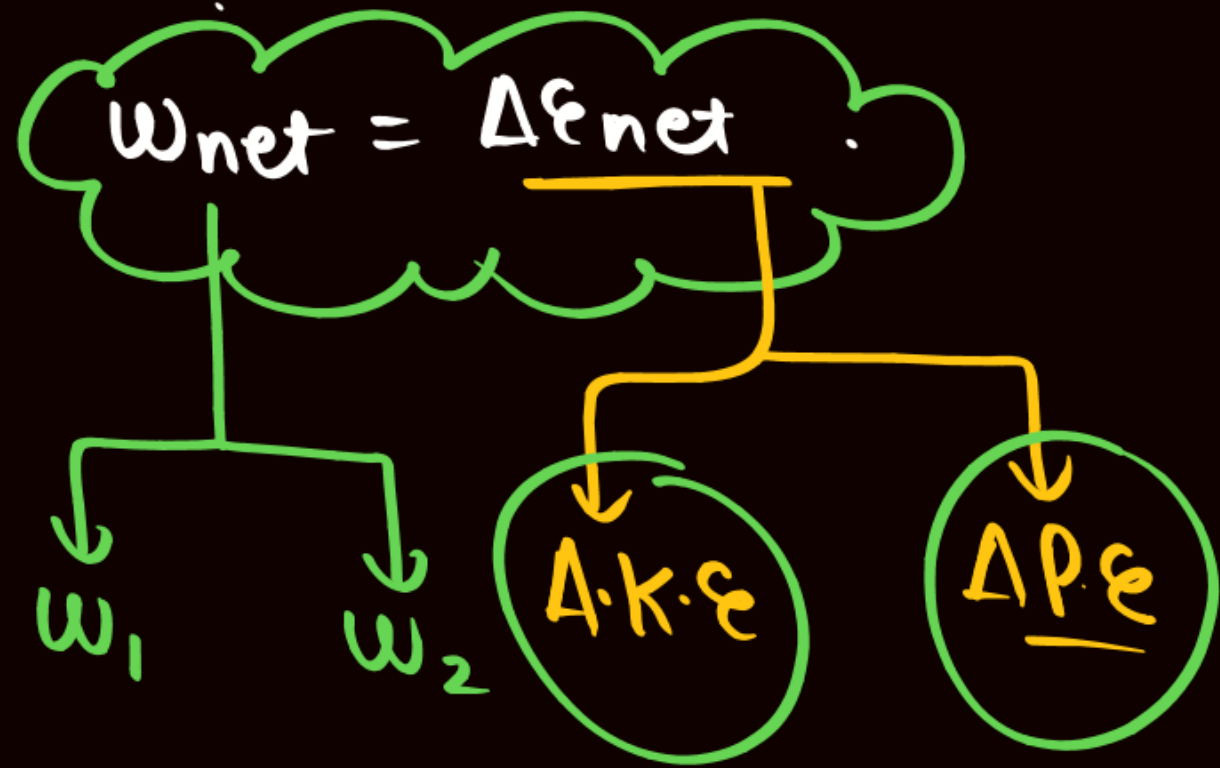
$$P_1 + \frac{\rho \cdot \mathcal{E}_1}{V} + \frac{\rho \cdot \mathcal{E}_1}{V} = P_2 + \frac{\rho \cdot \mathcal{E}_2}{V} + \frac{\rho \cdot \mathcal{E}_2}{V}$$



Derivation (B-Theorem)

$$\bullet W = \Delta E$$

work = change in energy



$$\text{WORK} = \Delta \cdot E$$

$$\text{WORK} = \Delta P \cdot E + \Delta K \cdot E$$

① $\Delta K.E$ (change in K.E)

$$\Delta K.E = K.E_f - K.E_i$$

$$\Delta K.E = K.E_2 - K.E_1$$

$$\Delta K.E = \frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2$$

$$\Delta K.E = \frac{1}{2} m (v_2^2 - v_1^2)$$

$$\Delta K.E = \frac{1}{2} \rho a v_1 t (v_2^2 - v_1^2)$$

Note

$$\rho = \frac{m}{V}$$

$$V \rho = m$$

$$m = V \rho$$

$$m = a_1 d_1 \rho$$

$$m = \rho a_1 d_1$$

$$m = \rho a_1 v_1 t$$

$$\text{Volume} = a_1 \times d_1$$

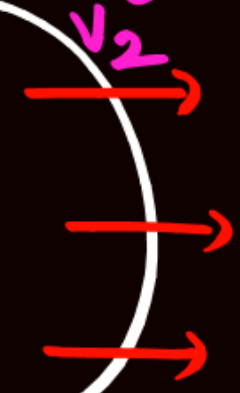
Velocity
initial



$$v_1 = \frac{d_1}{t}$$

$$v_1 t = d_1$$

Final



K.E2

$$v = a \times h$$

$$v = a_1 d_1$$

① $\Delta P.E$ (change in P.E)

$$\Delta P.E = P.E_f - P.E_i$$

$$\Delta P.E = P.E_2 - P.E_1$$

$$\Delta P.E = \underline{mgh_2} - \underline{mgh_1}$$

$$\Delta P.E = \underline{mg}(h_2 - h_1)$$

$$\Delta P.E = \rho a_1 v_1 t g (h_2 - h_1)$$

$$\rho = \frac{m}{V} \Rightarrow \text{volume } V = a_1 \times d_1$$

$$\rho V = m$$

$$m = \rho V$$

$$m = \rho a_1 \underline{d_1}$$

$$m = \rho a_1 v_1 t$$

velocity

initial



$$v_1 = \frac{d_1}{t}$$

$$v_1 t = d_1$$

Final

velocity

v_2

$$\underline{P.E_2 = mgh_2}$$

$P.E_2$

h_2

3 WORK done

$$w_{net} = w_1 + w_2$$

$$w_{net} = F_1 d_1 + -F_2 d_2$$

$$w_{net} = \underline{F_1} d_1 - \underline{F_2} d_2$$

$$w_{net} = P_1 a_1 \underline{d_1} - P_2 a_2 \underline{d_2}$$

$$w_{net} = P_1 a_1 v_1 t - P_2 \underline{a_2 v_2} t$$

$$w_{net} = \underline{P_1 a_1 v_1} t - \underline{P_2 a_1 v_1} t$$

$$w_{net} = a_1 v_1 t (P_1 - P_2)$$

Equation of continuity

$$a_1 v_1 = a_2 v_2$$

$$P_1 = \frac{F_1}{a_1}$$

$$P_1 a_1 = F_1$$

$$P_2 = \frac{F_2}{a_2}$$

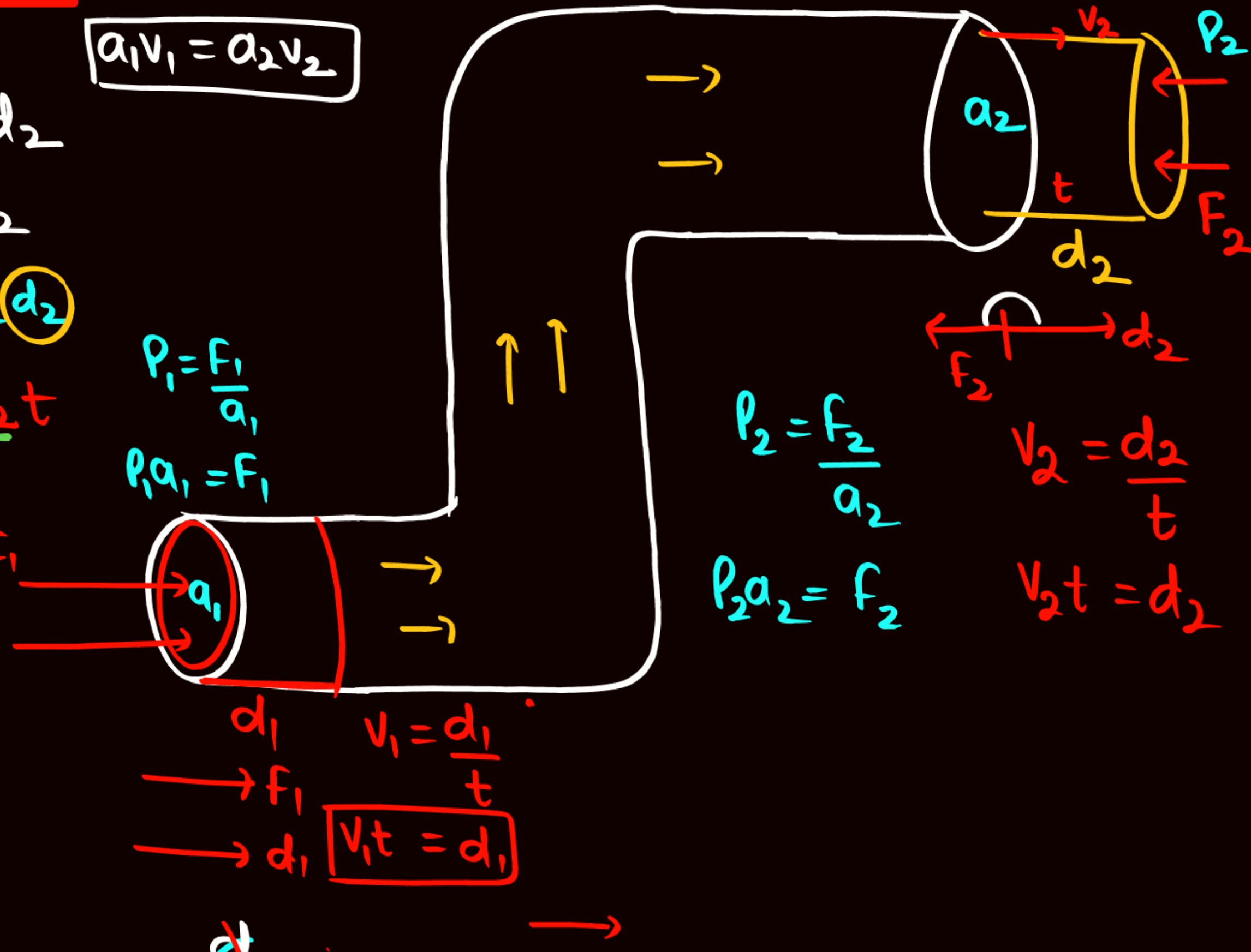
$$P_2 a_2 = F_2$$

$$v_2 = \frac{d_2}{t}$$

$$v_2 t = d_2$$

$$v_1 = \frac{d_1}{t}$$

$$v_1 t = d_1$$



$$\# W = \Delta K \cdot \epsilon + \Delta P \epsilon$$

$$\rho_1 v_1 t (P_1 - P_2) = \frac{1}{2} \rho_1 v_1 t (v_2^2 - v_1^2) + \rho_1 v_1 t g (h_2 - h_1)$$

$$\cancel{\rho_1 v_1 t} (P_1 - P_2) = \cancel{\rho_1 v_1 t} \left(\frac{1}{2} \rho (v_2^2 - v_1^2) + \rho g (h_2 - h_1) \right)$$

$$P_1 - P_2 = \frac{1}{2} \rho (v_2^2 - v_1^2) + \rho g (h_2 - h_1)$$

$$P_1 - P_2 = \frac{1}{2} \rho v_2^2 - \frac{1}{2} \rho v_1^2 + \rho g h_2 - \rho g h_1$$

$$P_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2$$

Bernoulli
Theorem

$$P_1 + \frac{1}{2} \frac{m}{V} v_1^2 + \frac{m}{V} g h_1 = P_2 + \frac{1}{2} \frac{m}{V} v_2^2 + \frac{m}{V} g h_2$$

$$P_1 + \frac{K \cdot \epsilon_1}{V} + \frac{P \cdot \epsilon_1}{V} = P_2 + \frac{K \cdot \epsilon_2}{V} + \frac{P \cdot \epsilon_2}{V}$$

$$\# \quad p_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = \uparrow p_2 + \frac{1}{2} \rho v_2^2 \downarrow + \rho g h_2$$

$$\textcircled{40} \quad \textcircled{40} \quad 20 \checkmark = \textcircled{60} + \textcircled{20} + 20 \checkmark$$

$$\epsilon_i = 100 \qquad \epsilon_f = 100$$

Special case



$$p_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = p_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2$$

$$p_1 + \frac{1}{2} \rho v_1^2 + \cancel{\rho g h_1} = p_2 + \frac{1}{2} \rho v_2^2 + \cancel{\rho g h_2}$$

$h_1 = h$ $h_2 = h$

$$p_1 + \frac{1}{2} \rho v_1^2 = p_2 + \frac{1}{2} \rho v_2^2$$

$$\downarrow 50 \quad 50 \quad 20 \quad 80 \uparrow$$

$$100 = 100$$

Note

$P \uparrow$

$K.E \downarrow \downarrow$

$K.E = \frac{1}{2}mv^2 \downarrow \downarrow$

$v \downarrow \downarrow$ (velocity)

$P \downarrow$

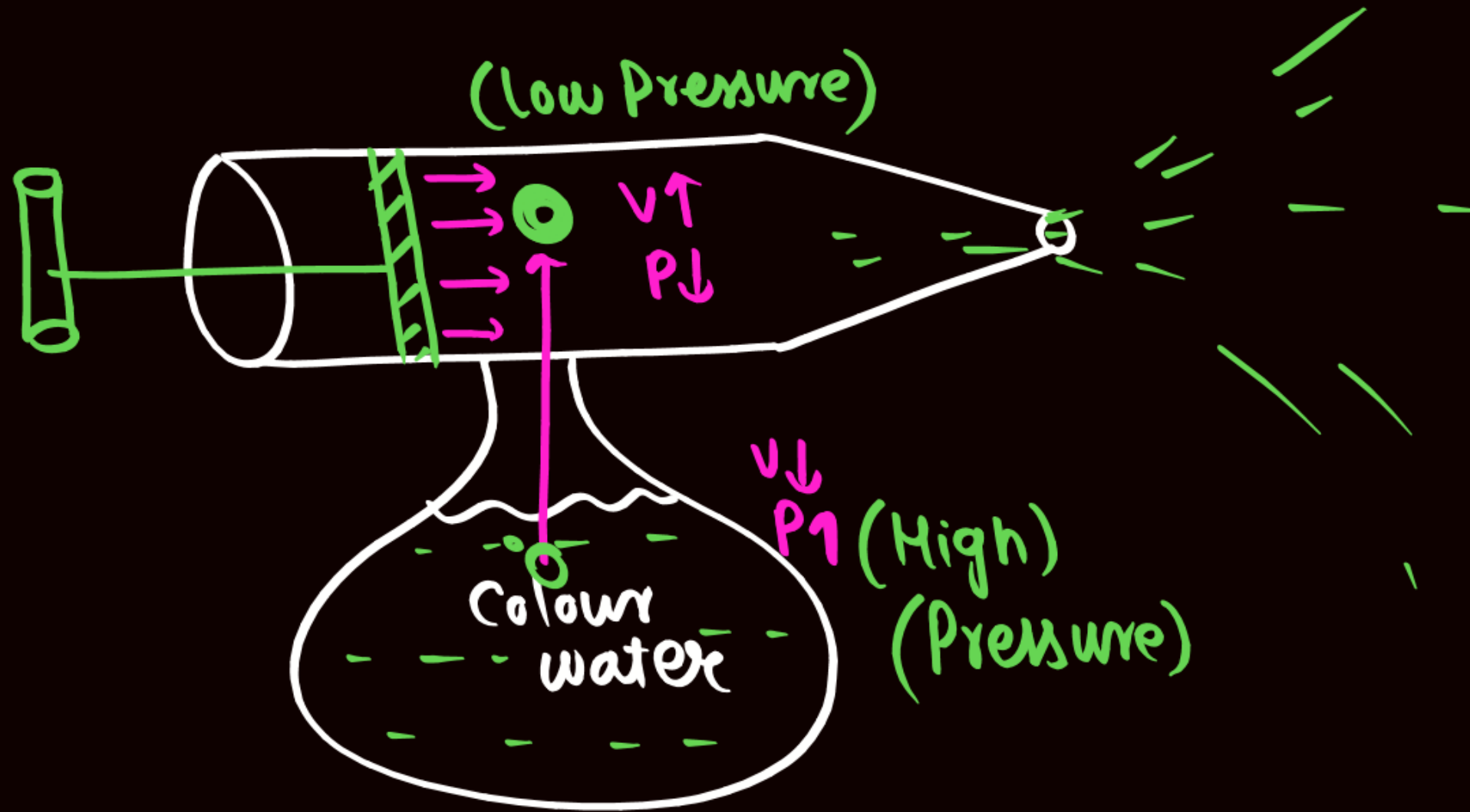
$K.E \uparrow$

$K.E = \frac{1}{2}mv^2 \uparrow \uparrow$

$v \uparrow \uparrow$ (velocity)

Application of Bernoulli's theorem ($P \uparrow V \downarrow$) ($P \downarrow V \uparrow$)

① SPRAY / Atomizer (Pichkarri)



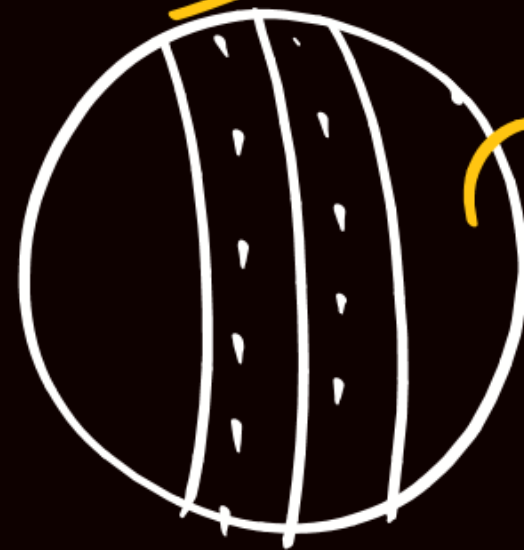
(ii) Magnus effect ← v_{swing}



Pressure $\uparrow$

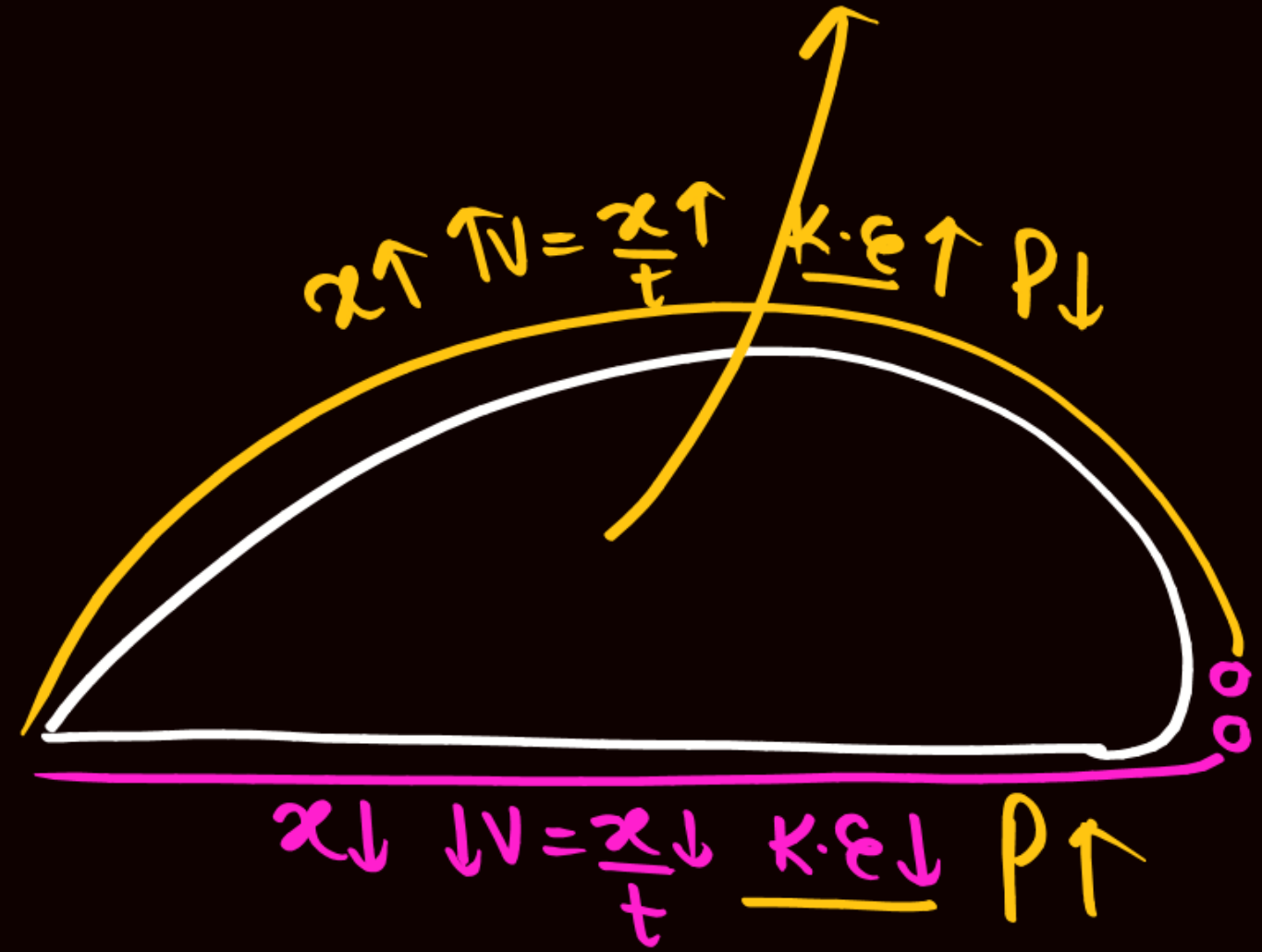
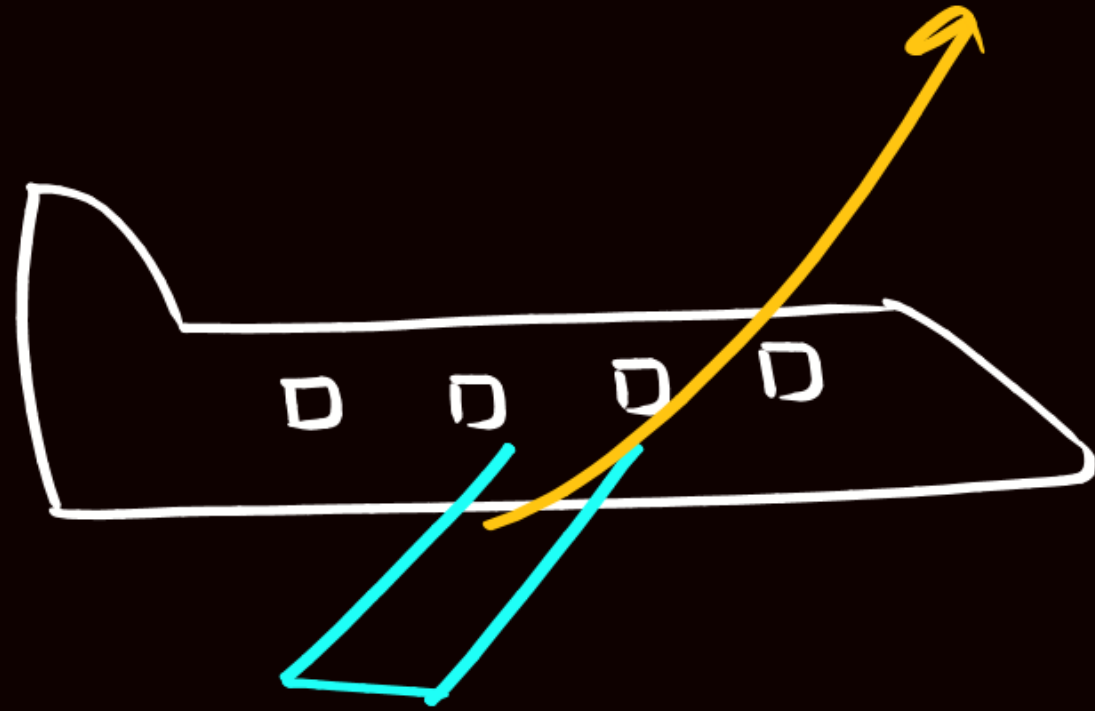
Outward

$P \uparrow$



air $V \uparrow$ $P \downarrow$

③ Flying of Aeroplane.



HOMework

Q Cp $\rightarrow$ derivation
x 2 times

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